

Requirements Specification

Tetris Game

SEDA Project Tutor

May 18, 2026

Contents

1	Document Control and Purpose	2
1.1	Project Goal	2
1.2	Stakeholders	2
2	Scope and Context	2
2.1	In Scope	2
2.2	Out of Scope	2
2.3	Assumptions and Dependencies	3
2.4	Glossary	3
3	Constraints	3
4	Functional Requirements	3
4.1	Use Cases	3
4.2	Detailed Functional Requirements	6
5	Quality Requirements	9
5.1	Usability	9
5.2	Performance	9
5.3	Reliability	9
5.4	Maintainability	10
6	Acceptance Criteria	10

1 Document Control and Purpose

This document provides the requirements specification for the two-player Tetris game, developed as part of the SEDA software engineering project. It describes the system requirements from the user and customer perspective, including local gameplay, LAN-based gameplay, player identification, interaction flow, gameplay behavior, and system constraints.

1.1 Project Goal

Deliver software that allows two users to play a Tetris game either on the same system through local gameplay or on separate systems connected through a local area network (LAN). The system shall provide smooth gameplay, clear board visibility, synchronized game state in LAN mode, player name support, and a consistent user experience across both play modes.

1.2 Stakeholders

- Players using the game in local or LAN mode.
- Customer requesting the development of the game.
- Developers implementing and maintaining the system.

2 Scope and Context

2.1 In Scope

- Local gameplay mode where two players play on the same system.
- LAN gameplay mode where two players play on separate systems through the same local network.
- Mode selection and game start flow.
- Player name handling for local and LAN gameplay.
- Block generation, movement, rotation, and falling behavior.
- Independent boards for both players.
- Inverted top board behavior.
- Line detection, line clearing, board update after clearing, and score update.
- Loss detection, continued gameplay, winner declaration, and restart behavior.
- Low-latency synchronization of board, score, player name, and game state information in LAN mode.

2.2 Out of Scope

- Internet-based multiplayer outside the local network.
- Mobile application version.
- Web-based version.
- AI opponent.

- Persistent user accounts, online leaderboard, or ranking system.

2.3 Assumptions and Dependencies

- The system runs on standard personal computers.
- In local mode, both players use the same system and separate keyboard controls.
- In LAN mode, both players use separate systems connected to the same local area network.
- Exactly two players participate in each game session.
- The system has access to keyboard input and graphical display.
- LAN communication is available before starting a LAN game.

2.4 Glossary

- **Block:** A falling Tetris shape controlled by the player.
- **Board:** The grid where blocks fall, stack, and form lines.
- **Line Clear:** Removal of a fully completed row from the board.
- **Local Mode:** Gameplay where both players use the same system.
- **LAN Mode:** Gameplay where players use separate systems connected through the same local area network.
- **Player Name:** A name or identifier used to identify each player during local or LAN gameplay.
- **Inverted Board:** The top player's board displayed upside down, where blocks move upward.
- **Game State:** Current board, score, active block, speed, player name, and player status.

3 Constraints

- **CON-TECH-01 (Technology):** The implementation language shall be Java version 25 or newer.
- **CON-PLAT-01 (Platform):** The system shall run on standard desktop or laptop computers.
- **CON-NET-01 (Network):** LAN gameplay shall operate only over a local area network.
- **CON-PLAY-01 (Players):** The system shall support exactly two players.
- **CON-INPUT-01 (Input):** The system shall support mouse and keyboard-based gameplay input.

4 Functional Requirements

4.1 Use Cases

UC-01 Start Local Game

- **ID:** UC-01

- **Description:** Start a two-player Tetris game session on the same system.
- **Actor:** Player
- **Preconditions:** Application is running and local mode is selected.
- **Postconditions:** Game starts and blocks begin to move on both boards.
- **Successful Course:** Players select local mode, the system assigns or accepts player names, initializes both boards and scores, and starts block movement.
- **Abnormal Course:** If the game cannot be initialized, the system shall not start the game and shall show an error message.
- **Priority:** High
- **Referenced Requirements:** FR-INPUT-01, FR-INPUT-02, FR-FLOW-01, FR-FLOW-03, FR-OUTPUT-01, FR-OUTPUT-02, FR-OUTPUT-03

UC-02 Start LAN Game

- **ID:** UC-02
- **Description:** Start a two-player Tetris game session on separate systems over LAN.
- **Actor:** Player
- **Preconditions:** Both systems are connected to the same local area network.
- **Postconditions:** LAN game starts and both systems show synchronized gameplay.
- **Successful Course:** Players select LAN mode, enter player names, the system shows available players, establishes communication, initializes both boards and scores, and starts synchronized block movement.
- **Abnormal Course:** If the LAN connection cannot be established, the system shall not start the game and shall show a connection error message.
- **Priority:** High
- **Referenced Requirements:** FR-INPUT-01, FR-INPUT-03, FR-INPUT-04, FR-FLOW-02, FR-FLOW-03, FR-FLOW-04, FR-OUTPUT-01, FR-OUTPUT-02, FR-OUTPUT-03, FR-OUTPUT-04

UC-03 Move and Rotate Block

- **ID:** UC-03
- **Description:** Move and rotate the active Tetris block during gameplay.
- **Actor:** Player
- **Preconditions:** Game is running and an active block is available.
- **Postconditions:** The active block position or orientation is updated.
- **Successful Course:** Player presses movement or rotation keys, and the system updates the active block if the move is valid.
- **Abnormal Course:** If the requested move is invalid because of a wall or another block, the system shall reject the move and keep the block in its current valid position.
- **Priority:** High
- **Referenced Requirements:** FR-INPUT-05, FR-FLOW-05, FR-FLOW-06, FR-FLOW-07, FR-OUTPUT-05, FR-OUTPUT-09

UC-04 Handle Block Falling

- **ID:** UC-04
- **Description:** Generate and move Tetris blocks during active gameplay.
- **Actor:** System
- **Preconditions:** Game is active and the player has not lost.

- **Postconditions:** Block moves toward its stacking position.
- **Successful Course:** The system generates a block, moves it downward on the bottom board or upward on the inverted top board, and locks it when it reaches a final position.
- **Abnormal Course:** If a new block cannot be placed due to board overflow, the system triggers the player loss condition.
- **Priority:** High
- **Referenced Requirements:** FR-FLOW-08, FR-FLOW-09, FR-FLOW-10, FR-FLOW-11, FR-FLOW-15, FR-OUTPUT-05

UC-05 Complete Line

- **ID:** UC-05
- **Description:** Detect and clear completed lines.
- **Actor:** System
- **Preconditions:** A row on a player board is completely filled.
- **Postconditions:** Completed line is cleared, blocks above the cleared line move downward, and score is updated.
- **Successful Course:** The system detects a completed line, removes the line, shifts all blocks above the cleared line downward to fill the empty space, and updates the player's score.
- **Abnormal Course:** If no completed line exists, the system keeps the board unchanged.
- **Priority:** High
- **Referenced Requirements:** FR-FLOW-12, FR-FLOW-13, FR-FLOW-14, FR-OUTPUT-06, FR-OUTPUT-07, FR-OUTPUT-09

UC-06 Player Loses

- **ID:** UC-06
- **Description:** Handle the condition where one player loses.
- **Actor:** System
- **Preconditions:** Blocks reach the losing threshold of one player's board.
- **Postconditions:** Losing player's score is stored and opponent continues playing.
- **Successful Course:** The system detects board overflow, stops block generation for the losing player, stores their score, and allows the opponent to continue.
- **Abnormal Course:** If both players lose at the same time, the system declares the winner based on the higher score or declares a draw if scores are equal.
- **Priority:** High
- **Referenced Requirements:** FR-FLOW-15, FR-FLOW-16, FR-FLOW-17, FR-FLOW-18, FR-OUTPUT-07, FR-OUTPUT-09

UC-07 End Game

- **ID:** UC-07
- **Description:** End the game and declare the winner.
- **Actor:** System
- **Preconditions:** One player has lost or the end condition has been reached.
- **Postconditions:** Winner is displayed and restart option is available.
- **Successful Course:** The system compares the game state and score, declares the winner, stops gameplay, and displays the restart button.
- **Abnormal Course:** If both scores are equal at the end condition, the system

declares a draw.

- **Priority:** High
- **Referenced Requirements:** FR-FLOW-17, FR-FLOW-18, FR-FLOW-19, FR-OUTPUT-07, FR-OUTPUT-08, FR-OUTPUT-10

UC-08 Restart Game

- **ID:** UC-08
- **Description:** Restart the game after it ends.
- **Actor:** Player
- **Preconditions:** Game has ended and restart button is displayed.
- **Postconditions:** Boards, scores, blocks, speed, player names, and player states are reset or restored for a new game.
- **Successful Course:** Player presses restart, and the system resets all game states and starts a new game session.
- **Abnormal Course:** If the LAN connection is lost in LAN mode, the system shall not restart the game and shall show a connection error message.
- **Priority:** Medium
- **Referenced Requirements:** FR-INPUT-06, FR-FLOW-02, FR-FLOW-03, FR-FLOW-20, FR-OUTPUT-08, FR-OUTPUT-10

4.2 Detailed Functional Requirements

ID	Description	Prio
FR-INPUT-01	The system shall allow users to select either local mode or LAN mode before starting the game.	High
FR-INPUT-02	The system shall assign default player names, such as Player 1 and Player 2, in local mode.	High
FR-INPUT-03	The system shall allow each player to enter a player name before starting a LAN game.	High
FR-INPUT-04	The system shall allow a LAN player to select an available opponent from the local network.	High
FR-INPUT-05	The system shall accept keyboard input for moving and rotating the active Tetris block.	High
FR-INPUT-06	The system shall allow the player to press a restart button after the game ends.	Medium
FR-FLOW-01	The system shall initialize a local two-player game session after local mode is selected.	High
FR-FLOW-02	The system shall establish communication between two player systems over the same local area network before LAN gameplay starts.	High
FR-FLOW-03	The system shall initialize independent game boards, scores, active blocks, speed, player names, and player states when a new game starts.	High
FR-FLOW-04	The system shall associate each board, score, and game state with the corresponding player name in LAN mode.	High

FR-FLOW-05	The system shall validate block movement before updating the block position.	High
FR-FLOW-06	The system shall validate block rotation before updating the block orientation.	High
FR-FLOW-07	The system shall reject invalid block movement or rotation and keep the block in the last valid position.	High
FR-FLOW-08	The system shall generate Tetris blocks continuously for each active player during gameplay.	High
FR-FLOW-09	The system shall move blocks downward on the bottom board and upward on the inverted top board.	High
FR-FLOW-10	The system shall lock a block into the board when it reaches its final valid position.	High
FR-FLOW-11	The system shall gradually increase the block movement speed during gameplay.	Medium
FR-FLOW-12	The system shall detect one or more completed horizontal lines on each player's board.	High
FR-FLOW-13	The system shall clear completed lines and update the board state by shifting all blocks above the cleared line or lines downward, ensuring that the empty spaces are filled.	High
FR-FLOW-14	The system shall update the player's score based on cleared lines.	High
FR-FLOW-15	The system shall detect when blocks reach the losing threshold of a player's board.	High
FR-FLOW-16	The system shall stop block generation for a player after that player loses.	High
FR-FLOW-17	The system shall store the score of the player who has lost.	High
FR-FLOW-18	The system shall allow the opponent to continue playing after one player loses until the win condition is reached.	High
FR-FLOW-19	The system shall determine the winner based on the final game state and player scores, or declare a draw if the scores are equal.	High
FR-FLOW-20	The system shall reset all game states when the restart option is selected.	High
FR-FLOW-21	In LAN mode, the system shall synchronize board state, score state, active player status, player names, and game result between both player systems during gameplay.	High
FR-FLOW-22	The system shall show a connection error or stop the LAN session safely if the LAN connection is lost.	Medium
FR-OUTPUT-01	The system shall display both players' boards simultaneously on the screen.	High
FR-OUTPUT-02	The system shall display the top player's board in inverted orientation.	High
FR-OUTPUT-03	The system shall display player names together with their respective boards and scores during gameplay.	High

FR-OUTPUT-04	In LAN mode, the system shall display a list of available players on the network using their player names.	High
FR-OUTPUT-05	The system shall visually update the active block position, orientation, and board state after each valid gameplay update.	High
FR-OUTPUT-06	The system shall display the updated board after completed lines are cleared and blocks above the cleared lines are shifted downward.	High
FR-OUTPUT-07	The system shall display both players' scores during gameplay.	High
FR-OUTPUT-08	The system shall display a restart button after the game ends.	Medium
FR-OUTPUT-09	In LAN mode, the system shall display synchronized board, score, and player status updates on both systems.	High
FR-OUTPUT-10	The system shall display the winner or draw result when the game ends.	High
FR-OUTPUT-11	There should be a part design option, which allows the user to specify own connected blocks.	High
FR-OUTPUT-12	During the game, objects should appear that, if hit, swap the players.	High
FR-OUTPUT-13	During the game, objects should appear that, if hit, increase the part speed of the opponent for a fixed amount of time.	High
FR-OUTPUT-14	During the game, objects should appear that, if hit, decreases the part speed, for a fixed amount of time.	High
FR-OUTPUT-15	During the game, objects should appear that, if hit, delays the opponent rotation input, for a fixed amount of time.	High
FR-OUTPUT-16	During the game, objects should appear that, if hit, delays the player's rotation input, for a fixed amount of time.	High
FR-OUTPUT-17	During the game, objects should appear that, if hit, decrease the speed for the opponent, for a fixed amount of time.	High
FR-OUTPUT-18	The speed of the game should be adjustable in the main menu.	High
FR-OUTPUT-19	If one line is cleared by one player, the screen should increase by one line and decrease by one line for the opponent.	High
FR-OUTPUT-20	There is an object that explodes on impact and destroys all block in a radius of 3 around the impact.	High

FR-OUTPUT-21	There is an object that explodes on impact and destroys all block below.	High
FR-OUTPUT-22	During the game, a portal should appear that takes the eliminates the block and forwards the block to the opponent as next appearing block.	High
FR-OUTPUT-23	During the game a teleporter object should appear, that swapes the blocks of the players.	High
FR-OUTPUT-24	There should be an advanced option that allows each player to control two blocks at once.	High
FR-OUTPUT-25	There should be the option to play horizontal instead of vertical.	High

5 Quality Requirements

5.1 Usability

- **QR-USAB-01:** The system shall allow users to clearly choose between local game-play and LAN gameplay.
- **QR-USAB-02:** The system shall provide simple keyboard controls for moving and rotating blocks.
- **QR-USAB-03:** The system shall clearly display both boards, player names, and both scores.
- **QR-USAB-04:** In LAN mode, the system shall make opponent selection understandable through player names.

5.2 Performance

- **QR-PERF-01:** In LAN mode, the system shall synchronize board and score updates between both players with low latency.
- **QR-PERF-02:** The system shall maintain smooth block movement during game-play.
- **QR-PERF-03:** The system shall update the board after a line clear within 1 second.

5.3 Reliability

- **QR-RELY-01:** The system shall complete a full two-player local game without crashing.

- **QR-RELY-02:** In LAN mode, the system shall keep both players' game states consistent during gameplay.
- **QR-RELY-03:** The scoring system shall remain consistent throughout a game session.
- **QR-RELY-04:** The system shall correctly update the board after completed lines are cleared.

5.4 Maintainability

- **QR-MAIN-01:** The project shall build with a documented command and provide reproducible run instructions.
- **QR-MAIN-02:** The implementation shall be structured into clearly separated components for input handling, gameplay flow, output display, and LAN communication.
- **QR-MAIN-03:** Automated tests shall cover mode selection, player names, game setup, movement validation, line clearing, score update, loss detection, LAN synchronization, and restart logic.

6 Acceptance Criteria

The Tetris game specification is considered acceptable when all high-priority functional and quality requirements are testable, two players can complete a full game in local mode, two players can complete a full game in LAN mode over the same local network, player names are correctly shown and associated with boards and scores, board and score states remain synchronized during LAN gameplay, completed lines are cleared with blocks above them shifted downward, and no listed constraint is violated.